

lecture

November 3, 2025

1 embedding

$\forall x, y, z, w \in X$

$$\varphi : X \rightarrow \mathbb{R}^d \quad (1)$$

ST $\delta(x, y) < \delta(w, z) \implies |\varphi(x) - \varphi(y)| < |\varphi(w) - \varphi(z)|$

there are $\binom{n}{2} = \theta(n^2)$ comparisons

there are $n!$ permutations of orderings on δ . each δ_{ij} has some value and any permutation of this is allowed as the ordering for δ .

2 dimension

lower bound is best case embedding, upper is worst case embedding (as functions of n).

the upper bound for n points is n . (remember this is *Ordinal*. this is not true for isometries) and lower bound $\frac{n}{2}$.

SEE LEMMA 2 in bilu linal

2.1 upper bound

take the ρ that gives rise to the ordering on δ_{ij} . take a symmetric matrix ϵ . diagonal zero, $\epsilon_{ij} < \frac{1}{n}$. where ϵ_{ij} are the consistent w permutations on ρ .

find a matrix $V = I - \epsilon$.

since it is diagonal dominant and symmetric, the matrix is positive definite (?)

thus there exists some M ST $MM^T = I - \epsilon$

for each u_i which is M_i (the vector of row i), we know that $u_i \cdot u_j = -\epsilon_{ij}$

but then

$$|u_i - u_j|^2 = |u_i|^2 - 2u_i u_j + |u_j|^2 = 2(1 + \epsilon_{ij}) \quad (2)$$

□

2.2 lower bound

$\forall c > 0$, and large enough n , almost no permutation ρ can be realized if $d < \frac{n}{2+c}$.

if the number of ordinal embeddings is less than the number of ρ on n points, then here must be an error (?)

consider a quadruplet i_1, j_1 vs i_2, j_2 define a poly $P_{i_1 j_1, i_2 j_2}(x) = |x_{i_1} - x_{j_1}|^2 - |x_{i_2} - x_{j_2}|^2$ depends on 4 things

- $\# \text{ vars} = n \cdot d$
- number of polys m , $m = O(n^4)$
- degree = 2

n is number of points to be embedded, d is dimension

note P encodes if i_1, j_1 is less than i_2, j_2

ordinal embeddings have 1 to 1 correspondance w polys here. each P encodes one comparison.

x has $n \cdot d$ variables.

give constructions of these graphs in linal